

Math Chapter 3: pt-2

① Uniform Continuous Distribution

X is uniformly distributed in $[a, b]$

$$f_x(x) = \begin{cases} 1/(b-a), & a \leq x \leq b \\ 0, & \text{o/t} \end{cases}$$

$$F_x(x) = \int_a^x 1/(b-a) dx$$

$$E[X] = (a+b)/2$$

$$\text{Var}[X] = (b-a)^2/12$$

② Exponential

X is the time for the first event in a Poisson RV to take place.

derivation

Let, $N(t)$ = no. of events during time t .

$$\hookrightarrow N(t) \sim \text{Poisson}(\lambda t)$$

X = waiting time until first event.

If waiting time before first event is greater than t ,
 $N(t) = 0$.

$$P[X > t] = P[N(t) = 0] = \frac{e^{-\lambda t} (\lambda t)^0}{0!} = e^{-\lambda t}$$

$$P[X > t] = e^{-\lambda t}$$

$$P[X \leq t] = F_X(t) = 1 - e^{-\lambda t}$$

$$F_X(t) = \begin{cases} 1 - e^{-\lambda t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$f_X(t) = \frac{d}{dt} F_X(t)$$

$$f_X(t) = \lambda e^{-\lambda t}$$

$$f_X(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$E[X] = 1/\lambda$$

$$\text{Var}[X] = 1/\lambda^2$$

③ Erlang

Time until n^{th} event in a Poisson RV to take place.

derivation

$N(t)$ = no. of events occurring by time t .

↳ $N(t) \sim \text{Poisson}(\lambda t)$

Let the time at which n^{th} event occurs be T_n .

If at a time t , $T_n \leq t \rightarrow n^{\text{th}}$ event already occurred at t .

$$\{T_n \leq t\} = \{N(t) \geq n\}$$

Let T_n be X

$$F_X(t) = P[X \leq t] = P(N(t) \geq n) = \sum_{i=n}^{\infty} \frac{e^{-\lambda t} (\lambda t)^i}{i!}$$

$$F_X(t) = \sum_{i=n}^{\infty} \frac{e^{-\lambda t} (\lambda t)^i}{i!}$$

$$f_x(t) = \sum_{i=n}^{\infty} \frac{d}{dt} \left[\frac{e^{-\lambda t} (\lambda t)^i}{i!} \right]$$

$$f_x(t) = \sum_{i=n}^{\infty} \left[\frac{-\lambda e^{-\lambda t} (\lambda t)^i}{i!} + \frac{i(\lambda t)^{i-1} (\lambda) e^{-\lambda t}}{i!} \right]$$

$$f_x(t) = \sum_{i=n}^{\infty} \left[\frac{-\lambda e^{-\lambda t} (\lambda t)^i}{i!} + \frac{\lambda e^{-\lambda t} (\lambda t)^{i-1}}{(i-1)!} \right]$$

$$f_x(t) = - \sum_{i=n}^{\infty} \frac{\lambda e^{-\lambda t} (\lambda t)^i}{i!} + \sum_{i=n}^{\infty} \frac{\lambda e^{-\lambda t} (\lambda t)^{i-1}}{(i-1)!}$$

$$f_x(t) = - \sum_{i=n}^{\infty} \frac{\lambda e^{-\lambda t} (\lambda t)^i}{i!} + \frac{\lambda e^{-\lambda t} (\lambda t)^{n-1}}{(n-1)!}$$

$$+ \sum_{i=n+1}^{\infty} \frac{\lambda e^{-\lambda t} (\lambda t)^{i-1}}{(i-1)!}$$

~~$f_x(t)$~~ Let $j = n + i - 1$

$$f_x(t) = - \sum_{i=n}^{\infty} \frac{\lambda e^{-\lambda t} (\lambda t)^i}{i!} + \frac{\lambda e^{-\lambda t} (\lambda t)^{n-1}}{(n-1)!}$$

$$\sum_{j=n}^{\infty} \frac{\lambda e^{-\lambda t} (\lambda t)^j}{j!}$$

$$f_x(t) = \frac{\lambda e^{-\lambda t} (\lambda t)^{n-1}}{(n-1)!}$$

$$E[X] = k/\lambda$$

$$\text{Var}[X] = k (\lambda)^{-2}$$

④ Gamma

gamma function

$$\textcircled{i} \quad \Gamma(x) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx, \quad x > 0$$

$$\textcircled{ii} \quad \Gamma(n) = (n-1)!, \quad n \in \mathbb{Z}$$

$$\textcircled{iii} \quad \Gamma(n) = (n-1) \Gamma(n-1), \quad n \in \mathbb{Z}$$

$$\textcircled{iv} \quad a) \Gamma(1) = 1$$

$$b) \Gamma(1/2) = \sqrt{\pi}$$

gamma distribution

$$f_x(x) = \begin{cases} \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

if $x \in \mathbb{Z}$

$$f_x(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{(r-1)!} \rightarrow \text{Erlang PDF}$$

$\therefore$ Erlang is a gamma distribution with an integer shape parameter

$$E[X] = r/\lambda$$

$$\text{Var}[X] = r/\lambda^2$$

⑤ Beta

beta function

$$① \quad B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$② \quad B(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

beta distribution

$$f_x(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1 \\ 0, & \text{o/t.} \end{cases}$$

$$E[X] = \frac{\alpha}{\alpha+\beta}$$

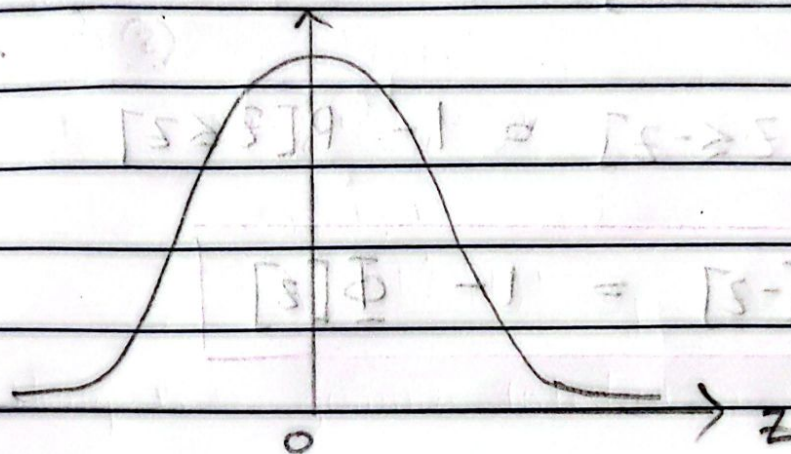
$$\text{Var}[X] = \frac{\alpha\beta}{(\alpha+\beta)^2 (\alpha+\beta+1)}$$

⑥ Gaussian / Normal RV $N(\mu, \sigma^2)$

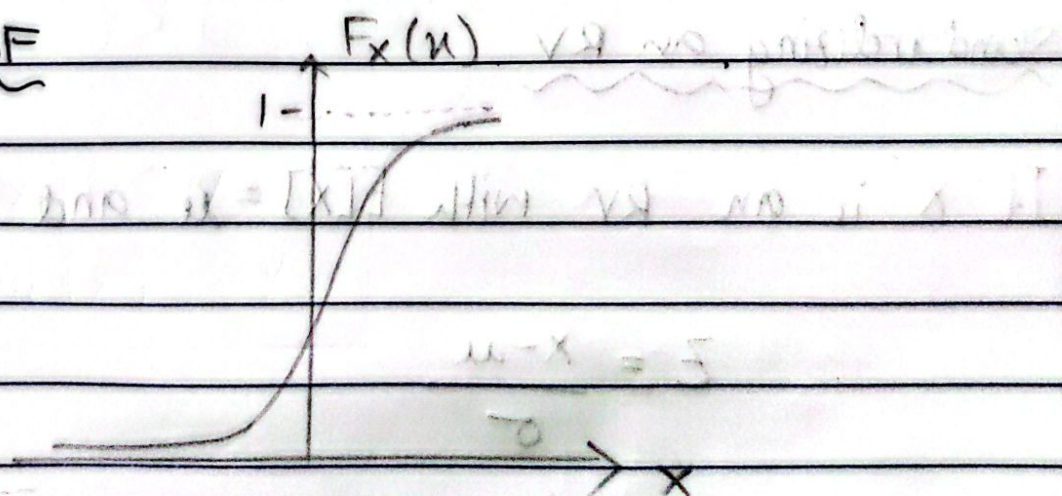
$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \times e^{\left[\frac{-(x-\mu)^2}{2\sigma^2} \right]}$$

$$F_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \times \int_{-\infty}^{+\infty} e^{\left[\frac{-(x-\mu)^2}{2\sigma^2} \right]}$$

PDF



CDF



$$\Phi(z) = P[Z \leq z]$$

Since it is symmetric,

$$P[Z \leq -z] = P[Z \geq z]$$

hence, $\Rightarrow P[Z \geq z] = 1 - P[Z \leq z]$

~~$$P[Z \geq z] = 1$$~~

$$\therefore P[Z \leq -z] = 1 - P[Z \leq z]$$

$$\Phi[-z] = 1 - \Phi[z]$$

Standardizing an RV

If X is an RV with $E[X] = \mu$ and $\text{Var}[X] = \sigma^2$,

$$Z = \frac{X - \mu}{\sigma}$$

is a standard / normal RV with $E[Z] = 0$
and $\text{Var}[Z] = 1$.

$$E[z] = E\left[\frac{X - \mu}{\sigma}\right]$$

$$= \frac{1}{\sigma} E[X - \mu]$$

$$= \frac{1}{\sigma} (E[X] - \mu)$$

$$= \frac{1}{\sigma} (\mu - \mu)$$

$$E[z] = 0$$

$$\text{Var}[z] = \text{Var}\left[\frac{X - \mu}{\sigma}\right]$$

$$= \frac{1}{\sigma^2} \text{Var}[X - \mu]$$

$$= \frac{1}{\sigma^2} \text{Var}[X]$$

$$= \frac{1}{\sigma^2} \times \sigma^2$$

$$\text{Var}[z] = 1$$

⑦ Lognormal Distribution lognormal (μ, σ^2)

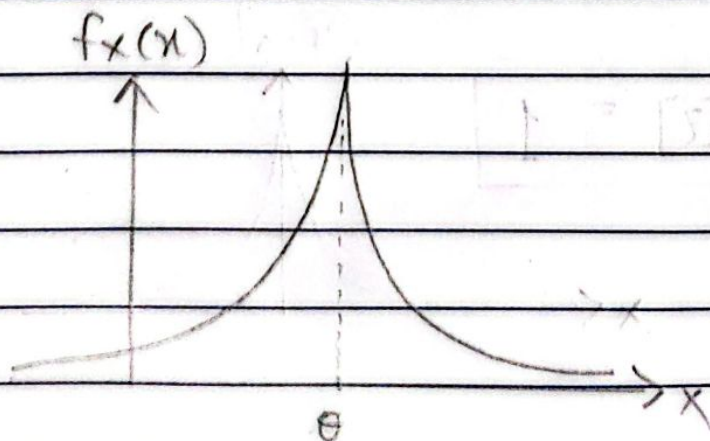
A positive RV X is lognormally distributed when taking its natural logarithm produces a normal RV.

$$\ln X \sim N(\mu, \sigma^2)$$

⑧ Laplace Distribution Laplace (θ, σ)

$$f_X(x) = \frac{1}{2\sigma} e^{-\frac{|x-\theta|}{\sigma}} \quad \theta = \text{centre of the distrib.}$$

$$f_X(x) = \begin{cases} \frac{1}{2\sigma} e^{(x-\theta)/\sigma} & x < \theta \\ \frac{1}{2\sigma} e^{-(x-\theta)/\sigma} & x \geq \theta \end{cases}$$



Chapter 3 : part 3

① Multivariate Hypergeometric RV

like picking balls from an urn without replacement.

r : different types

n_i : number of i th type selected/picked. present.

x_i : number of i th type picked.

$$P_{X_1, X_2, \dots, X_r} (x_1, \dots, x_r) = \frac{\binom{n_1}{x_1} \binom{n_2}{x_2} \dots \binom{n_r}{x_r}}{\binom{n_1 + n_2 + \dots + n_r}{n}}$$

$\rightarrow n_1 + n_2 + \dots + n_r = n$

② Multinomial Distribution

like picking balls from an urn with replacement

k : no. of diff types

n_i : no. of i th type present

x_i : no. " " " picked.

$$P_{X_1, X_2, \dots, X_k} (x_1, x_2, \dots, x_k) = \frac{n!}{x_1! x_2! \dots x_k!} \times p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

where $p_i = \frac{n_i}{n}$; $x_1 + x_2 + \dots + x_k = n$

Categorical Distribution

Like picking a target ball from an urn.

Bivariate Gaussian (Normal) Distribution

Standard distribution in a bivariate RV.

Chapter 4 : part 1

① Sum Rule / Law of Total Probability

$$(a) P_X(x) = \sum_y P_{X|Y}(x|y) \times P_Y(y)$$

$$(b) P_X(x) = \sum_y P_{X,Y}(x,y)$$

② Product Rule

$$P_{X,Y}(x,y) = P_{Y|X}(y|x) P_X(x)$$

③ Conditional Expectation

$$\text{if } E[X] = \sum_x x P_X(x),$$

$$\text{then } E[X|A] = \sum_x x P_{X|A}(x|a)$$

④ Law of Total ~~Expectation~~ Expectation

$$(a) E[X] = \sum_y E[X|Y=y] P_Y(y)$$

$$(b) E[Y] = \sum_x E[Y|X=x] P_X(x)$$

Bayes Theorem for Discrete RV

$$P_{Y|X}(y|x) = \frac{P_{Y|X}(y|x) P_X(x)}{\sum_x P_{Y|X}(y|x) P_X(x)}$$

$$\Rightarrow P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$$

Conditioned on Another RV

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

Chapter 4: part 3 <<Discrete>>

① Independence of functions

If X and Y are independent,

$$a) P[X=x, Y=y] = P[X=x] \times P[Y=y]$$

$$b) P[X=x | Y=y] = P[X=x]$$

② Independence on derived functions & expected values

If X and Y are independent,

$$a) E[XY] = E[X]E[Y] \text{ \& } \text{Cov}(X, Y) = 0.$$

However, $E[XY] = E[X]E[Y]$ simply does not prove the independence of X and Y , it merely proves that $\text{Cov}(X, Y) = 0$.

$$b) E[g(X)h(Y)] = E[g(X)]E[h(Y)] \text{ if } X \text{ and } Y \text{ are independent}$$

<< Continuous >>

① Independence of functions

If X and Y are independent,

$$a) f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

$$b) f_{X|Y}(x|y) = f_X(x)$$